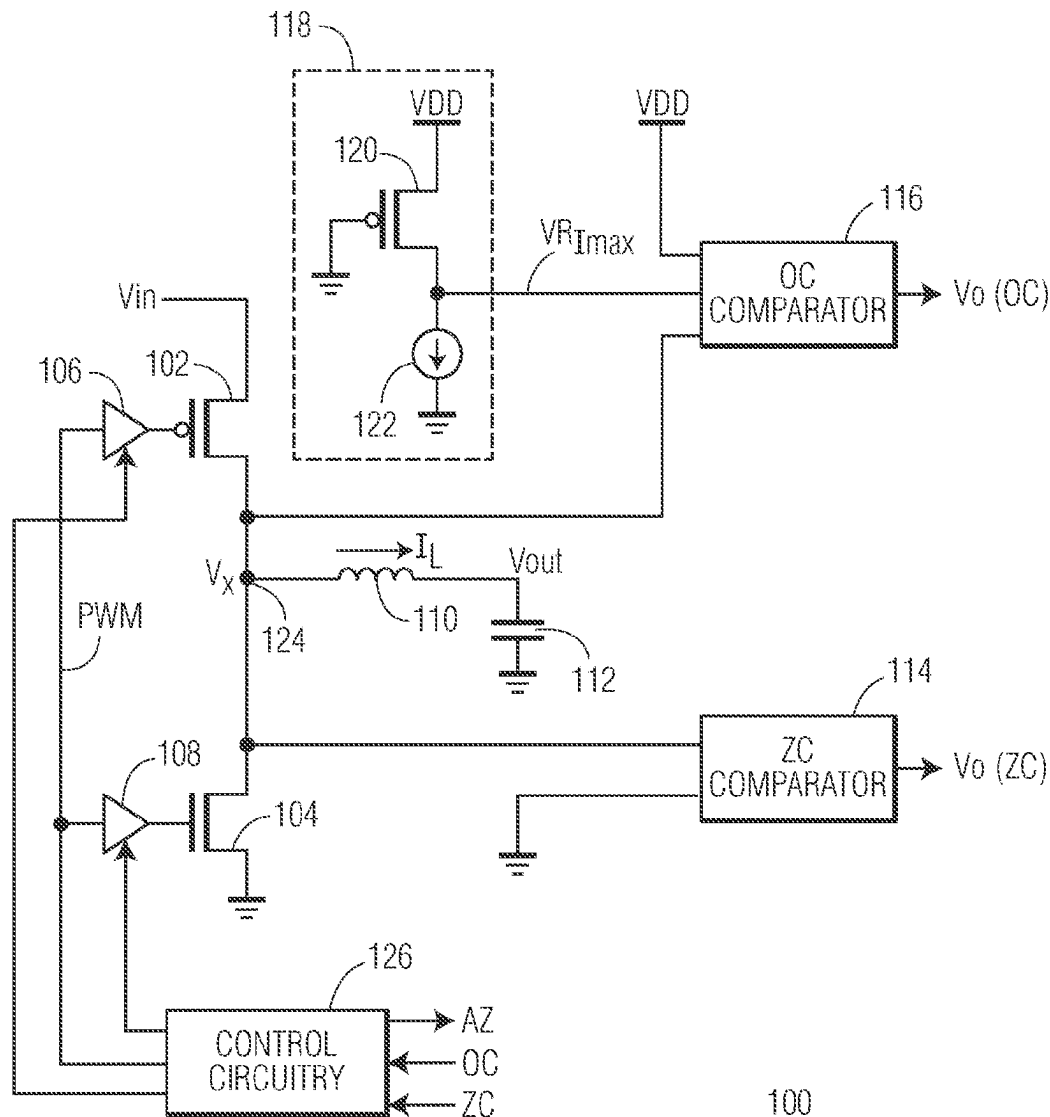




US 20250277824A1

(19) **United States**(12) **Patent Application Publication**
Mertens(10) **Pub. No.: US 2025/0277824 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **CURRENT COMPARATOR USABLE IN
DC-DC CONVERTER APPLICATIONS**(71) Applicant: **NXP B.V.**, Eindhoven (NL)(72) Inventor: **Robert Matthew Mertens**, Crystal
Lake, IL (US)(21) Appl. No.: **18/591,341**(22) Filed: **Feb. 29, 2024****Publication Classification**(51) **Int. Cl.**
G01R 19/00 (2006.01)
H02M 1/00 (2007.01)
H02M 3/158 (2006.01)(52) **U.S. Cl.**
CPC **G01R 19/0038** (2013.01); **H02M 1/0009**
(2021.05); **H02M 3/158** (2013.01)(57) **ABSTRACT**

A current comparator includes a first capacitor having a first terminal coupled to a first input signal, a second capacitor having a first terminal coupled to a second input signal, a first transistor having a control electrode coupled to a second terminal of the first capacitor, and a third transistor having a control electrode coupled to a second terminal of the second capacitor. First current electrodes of the first and second transistors are coupled, and second current electrodes of the first and second transistors are coupled to first and second circuit nodes, respectively. A single-ended cascode amplifier has an input coupled via a third capacitor to the second circuit node. A set of auto-zero switches, in response to an auto-zero control signal, selectively shorts the control electrode and second current electrode of the first transistor and selectively shorts the control electrode and the second current electrode of the second transistor.



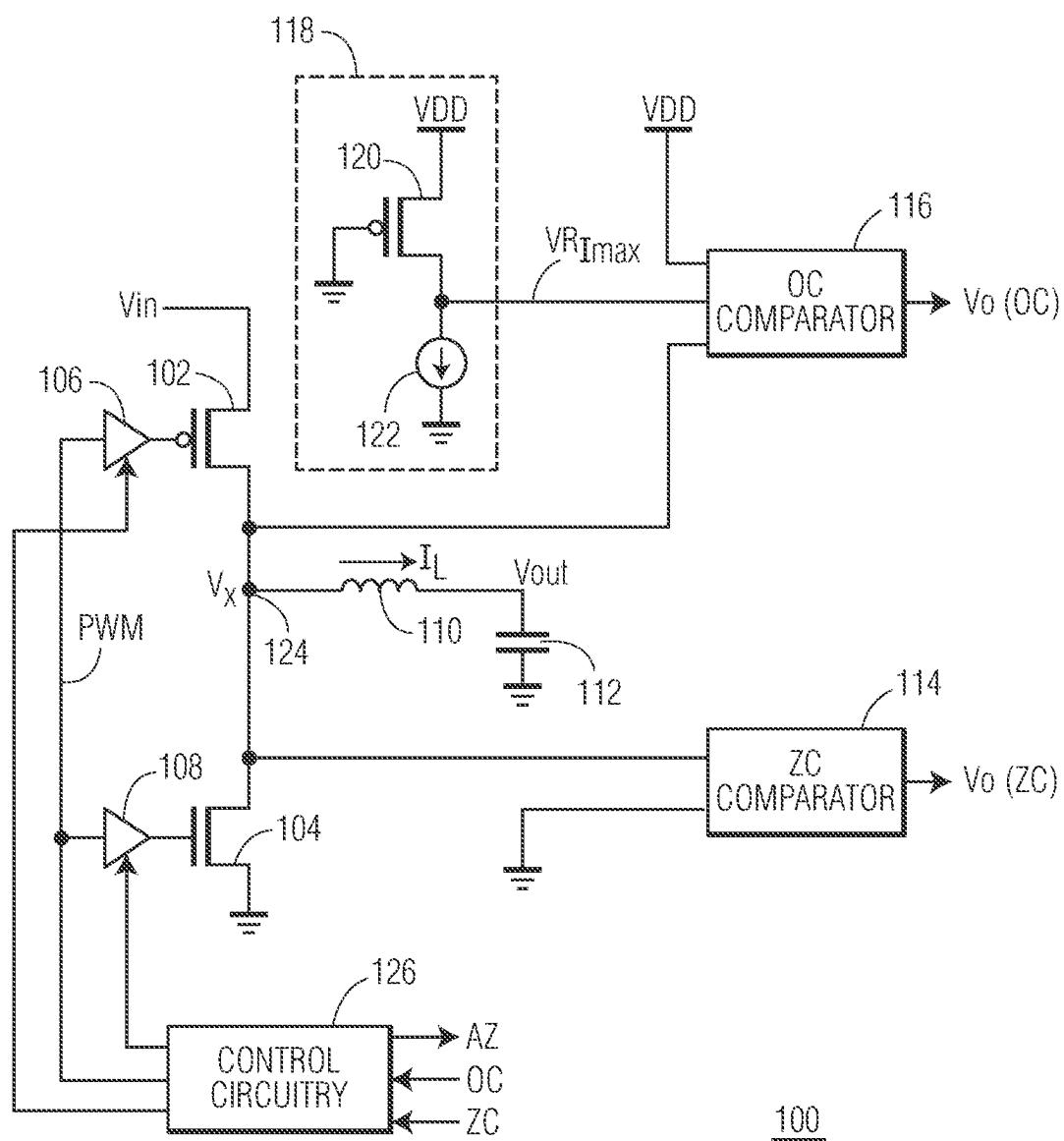


FIG. 1

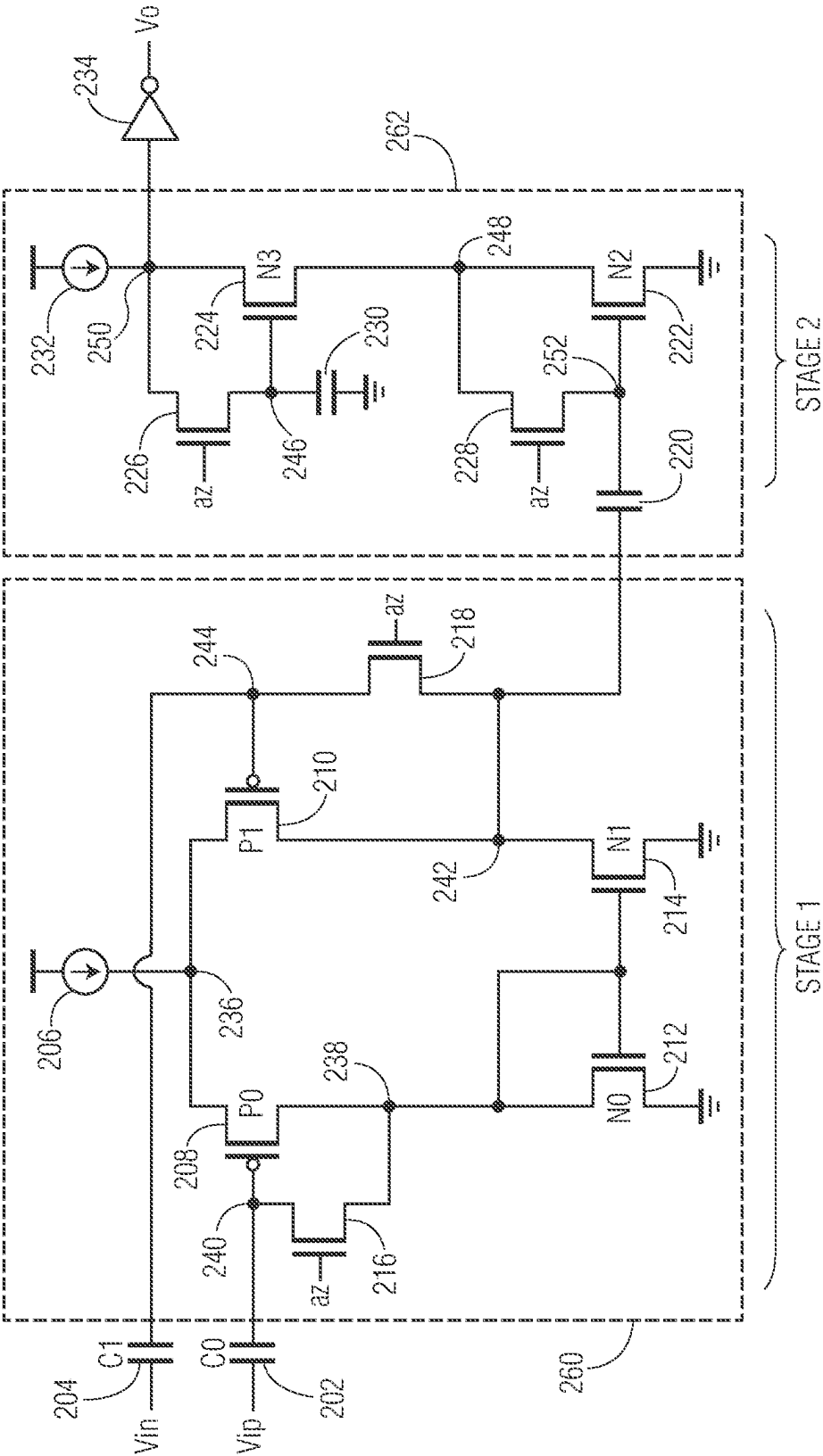
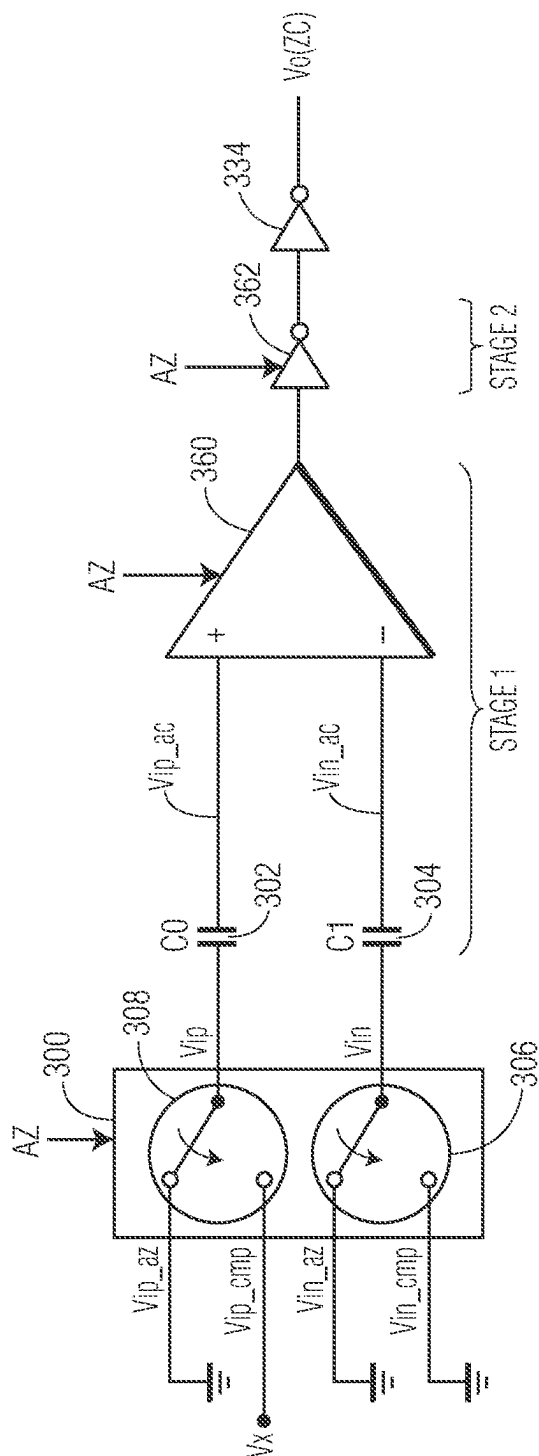
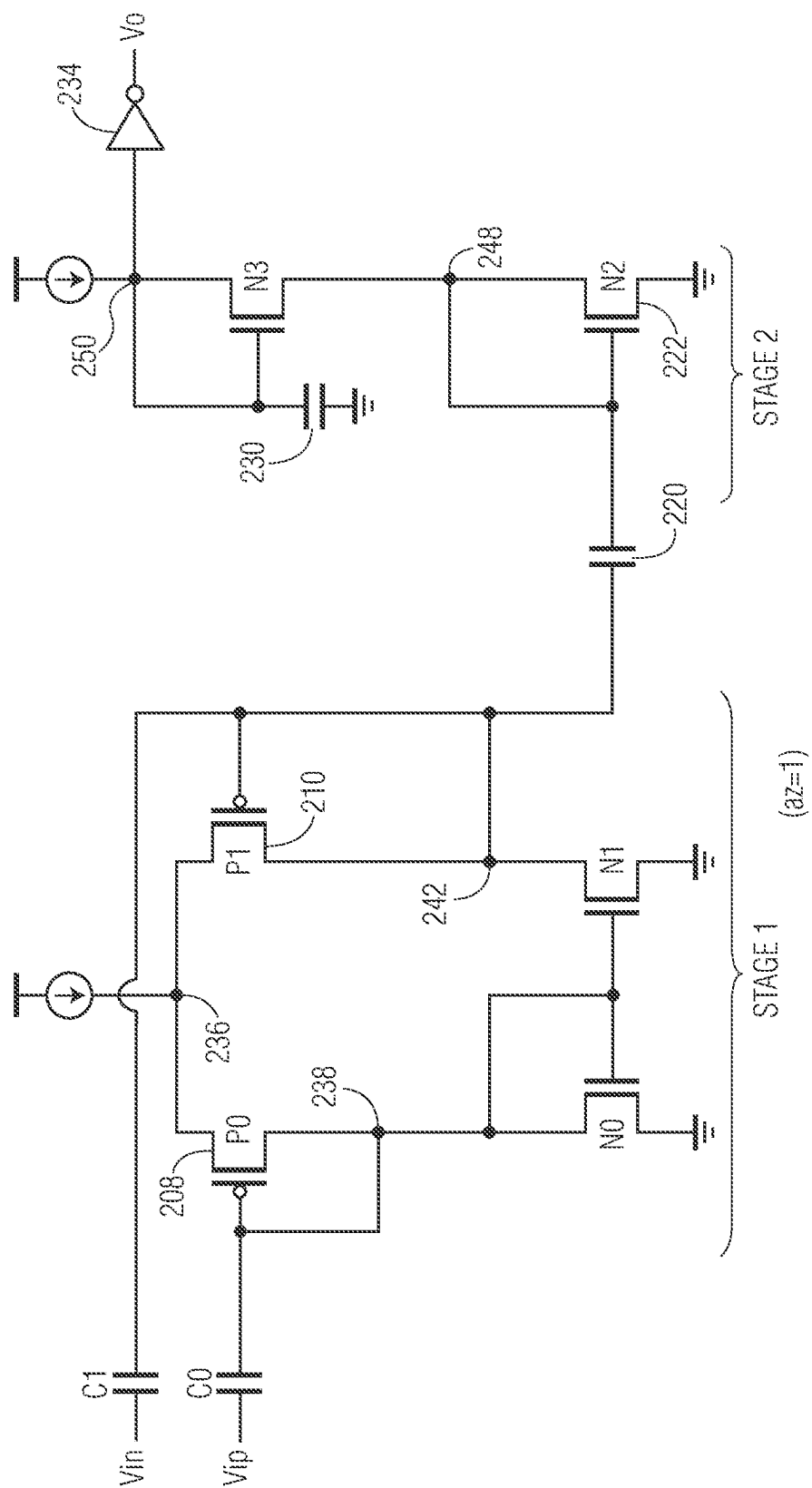


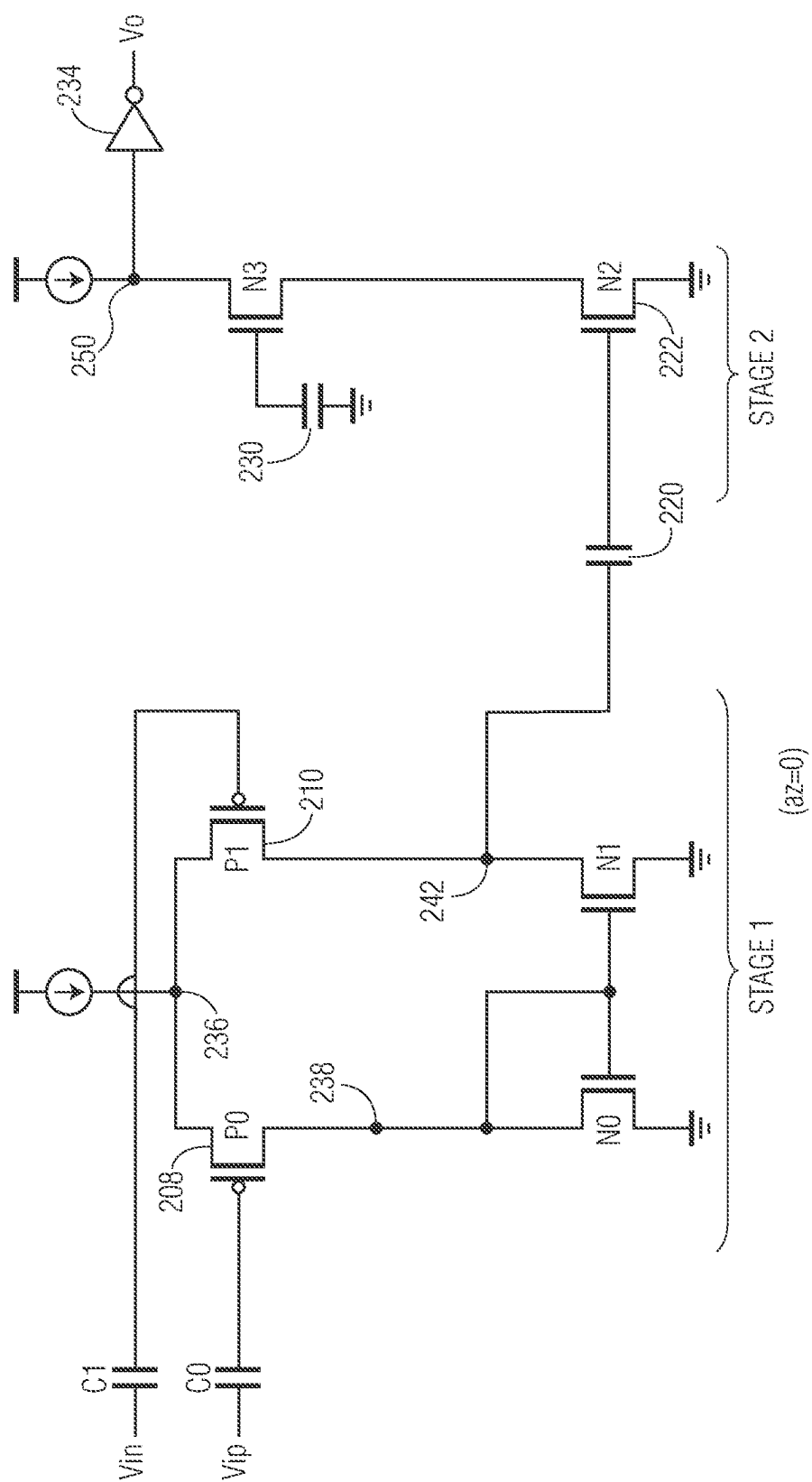
FIG. 2



354



ॐ
ॐ
ॐ



55

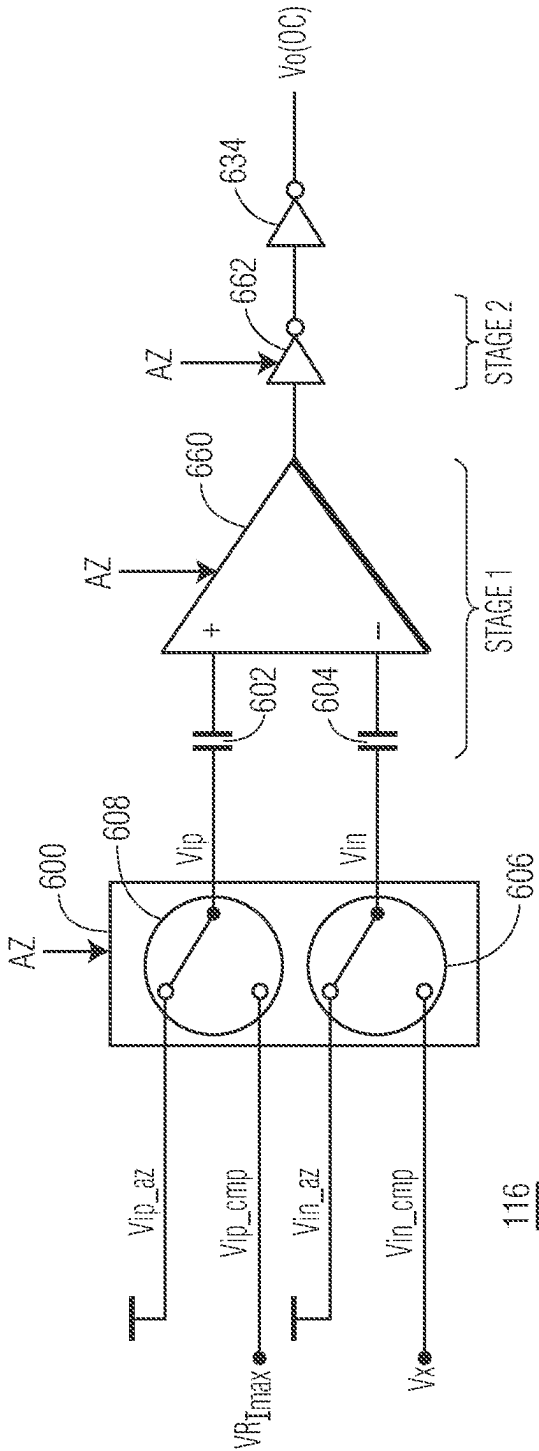


FIG. 6

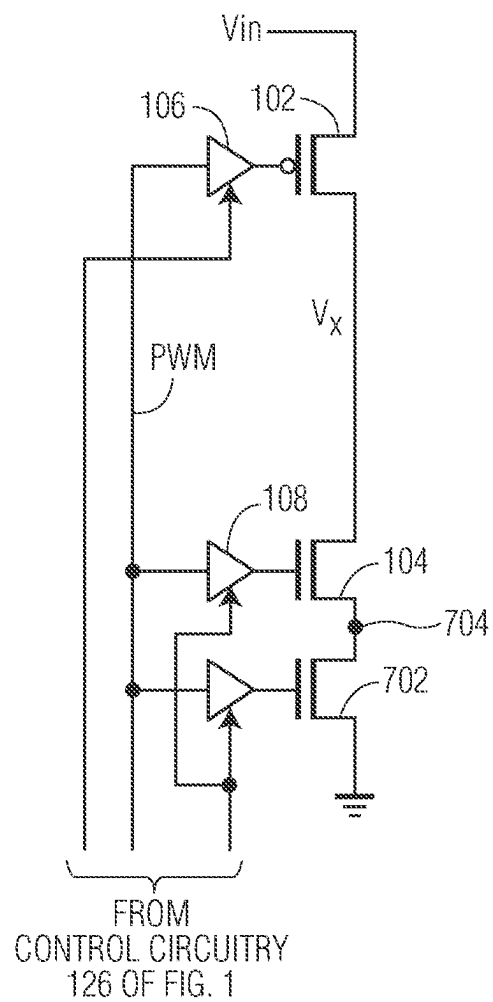


FIG. 7

CURRENT COMPARATOR USABLE IN DC-DC CONVERTER APPLICATIONS

BACKGROUND

Field

[0001] This disclosure relates generally to integrated circuits, and more specifically, to a current comparator usable in DC-DC converter applications.

Related Art

[0002] Direct current (DC)-direct current (DC) converters typically need to sense the current in the power transistors so as to avoid exceeding maximum current ratings or to sense when the inductor current crosses zero so as to support discontinuous conduction mode (DCM) operation. Therefore, a need exists for improved comparators capable of sensing such thresholds with improved precision while maintaining power conversion efficiency.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The present invention is illustrated by way of example and is not limited by the accompanying figures, in which like references indicate similar elements. Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale.

[0004] FIG. 1 illustrates a DC-DC buck converter having an over-current (OC) comparator and a zero-crossing (ZC) comparator, in accordance with one embodiment of the present invention.

[0005] FIG. 2 illustrates a current comparator, responsive to an auto-zero (AZ) control signal, which can be implemented by the OC comparator or ZC comparator of FIG. 1, in accordance with one embodiment of the present invention.

[0006] FIG. 3 illustrates a current comparator having a front-end input signal switch usable within the ZC comparator of FIG. 1 which is implemented as illustrated in FIG. 2, in accordance with one embodiment of the present invention.

[0007] FIGS. 4 and 5 illustrate the current comparator of FIGS. 2 and 3 during different phases of operation, corresponding to values of the AZ control signal, in accordance with one embodiment of the present invention.

[0008] FIG. 6 illustrates a current comparator having a front-end input signal switch usable within the OC comparator of FIG. 1 which is implemented as illustrated in FIG. 2, in accordance with one embodiment of the present invention.

[0009] FIG. 7 illustrates, in schematic form, a power stage of the DC-DC buck converter of FIG. 1 in accordance with one embodiment of the present invention.

DETAILED DESCRIPTION

[0010] In one aspect, a current comparator (also referred to as a threshold comparator) is provided which can implement a ZC comparator to indicate when the inductor current of a DC-DC converter crosses zero. In this manner, DCM operation of the DC-DC converter can be supported. The current comparator can also implement an OC comparator to indicate when the inductor current exceeds a maximum current rating. The current comparator includes a first stage (i.e. a differential stage) with input capacitors configured to store

offsets and power switches which enable auto-zeroing. The current comparator also includes a second stage including one or more single-ended cascode amplifiers in a cascaded configuration, each with a DC blocking capacitor on the input as well as a capacitor to store the cascode transistor's gate bias voltage and accompanying power switches to support auto-zeroing. The current comparator may also include output logic gates to convert the last single-ended cascode amplifier's analog output to a digital signal.

[0011] FIG. 1 illustrates, in partial block diagram and partial schematic form, a DC-DC converter **100** having an over-current (OC) comparator **116** (which may also be referred to as an OC detector or an OC condition detector) and a zero-crossing (ZC) comparator **114** (which may also be referred to as a ZC detector or a ZC condition detector), in accordance with one embodiment of the presents invention. DC-DC converter **100** is configured as a buck or step-down DC-DC converter which decreases voltage while increasing current from its input (V_{in}) to its output (load). In the illustrated embodiment, converter **100** is a switching converter, with power switches **102** and **104**, implemented as a p-channel metal-oxide semiconductor (PMOS) transistor and n-channel metal-oxide semiconductor (NMOS) transistor, respectively. Such switching converters typically provide for greater power efficiency as compared to linear regulators (which do not typically step up output current). (Note that, in alternate embodiments, each power switch can be implemented with multiple transistors, in which the power switches may be referred to collectively as the power stage of DC-DC converter **100**.) In addition to the power stage (e.g. power switches **102** and **104**), converter **100** includes energy storage elements inductor **110** and capacitor **112**. In the illustrated embodiment, a first current electrode of PMOS transistor **102** is coupled to receive an input voltage, V_{in} , and a second current electrode is coupled to a first circuit node **124** at a first terminal of inductor **110**. The voltage at circuit node **124** is labeled V_x (corresponding to a voltage at a first internal node of the power stage of converter **100**). In one embodiment, V_{in} may be coupled to a first voltage supply terminal configured to provide a first supply voltage, V_{dd} . A first current electrode of NMOS transistor **104** is coupled circuit node **124** and the second current electrode of transistor **102**, and has a second current electrode coupled to a second voltage supply terminal configured to supply a second supply voltage, V_{ss} , in which V_{dd} is greater than V_{ss} . In one embodiment, V_{ss} is ground, such as, e.g., a local AC ground. Note that, as used herein, the first and second voltage supply terminals may simply be referred to by the supply voltage provided by the terminal (e.g. as V_{dd} and V_{ss} , respectively). A second terminal of inductor **110** is coupled to a first terminal of capacitor **112** and provides an output voltage, V_{out} , of the converter, and a second terminal of capacitor **112** is coupled to V_{ss} . A control electrode of transistor **102** is coupled to an output of a first driver **106**, and a control electrode of transistor **104** is coupled to an output of a second driver **108**.

[0012] In the illustrated embodiment, an input of driver **106** and an input of driver **108** are each coupled to receive a digital input signal, such as a pulse-width modulated (PWM) signal which controls transistors **102** and **104** (also referred to as switches or power switches) to be in an on-state (in which transistor **102** is on while transistor **104** is off) or an off-state (in which transistor **102** is off but transistor **104** is on). That is, the PWM signal results in

transistors **102** and **104** being alternately on and off. Note that in alternate embodiments, different digital signals may be used for drivers **106** and **108** to control the on and off states of each of transistors **106** and **108** and thus control the generation of V_x . In one embodiment, the PWM may be provided by control circuitry **126**. Alternatively, it can be provided by circuitry outside converter **100**.

[0013] In continuous conduction mode (CCM) operation, the current through inductor **110** (indicated by the arrow labeled I_L) is controlled by power switches **102** and **104**, as known in the art, in which V_x at node **124** forms a square wave between a high voltage and a low voltage, in which the high voltage is between V_{dd} and the low voltage and the low voltage normally slightly below V_{ss} (alternatively, it may be slightly above V_{ss} if the converter permits I_L to be negative). For example, while in the on-state (during a low phase of the PWM signal which causes transistor **102** to be on but **104** to be off), V_x is at the high voltage in which I_L increases as inductor **110** charges, and while in the off-state (during a high phase of the PWM signal which causes transistors **102** to be off and **104** on), V_x is at the low voltage in which I_L decreases as inductor **110** discharges. In this manner, V_{out} is provided as an average level of V_x , between the high and low voltages of V_x , in which the LC filter formed by inductor **110** and capacitor **112** operates to flatten out V_{out} .

[0014] While in the on-state, if inductor **110** saturates, I_L may increase to a level beyond a maximum current rating of the inductor (referring to as I_{max}). If this happens, the buck converter may fail to operate correctly and can result in a reliability problem for the buck converter or any load driven by the buck converter. Therefore, as will be described below, OC comparator **116** is coupled circuit node **124** in order to monitor current through transistor **102** which corresponds to I_L through inductor **110** (since transistor **104** is off during the on-state). When I_L surpasses the predetermined maximum current, I_{max} , OC at the output of OC comparator **116** is asserted. OC can be provided to control circuitry **126** which disables (i.e. turns off) transistor **102** in response to assertion of OC (i.e. in response to OC indicating an over-current condition for I_L). That is, so long as an OC condition exists, control circuit **126** can maintain transistor **102** disabled. OC comparator **116** can be implemented using a current comparator which will be described below in reference to FIG. 2.

[0015] When converter **100** operates in continuous conduction mode, one of the two power switches is always on, so I_L is always rising or falling. However, in some cases, it may be preferable to disable both power switches at some times. For example, if the power required by the load is very small, it may be less efficient to operate in CCM with I_L swinging between positive and negative currents each cycle than it would be to operate in DCM. In DCM, first transistor **102** is turned on to increase I_L , then transistor **102** is turned off and transistor **104** is turned on so that I_L decreases, and when I_L reaches zero, transistor **104** is turned off, causing node V_x **124** to become high impedance (Z) and follow the voltage on V_{out} . With the same average I_L , the DCM will have lower RMS current than the CCM, and will often make converter **100** more efficient. As a second example, converter **100** might have an optimum efficiency at a specific output current in CCM. If such an optimum current exists, when the load current is below the optimum current, it may be most efficient to alternatively drive a current burst at the optimal output current and let I_L idle at zero so that the

converter's output current matches the current drawn by the load. In both examples, to avoid dissipating energy stored in inductor **110**, transistor **102** is kept on until I_L decreases to zero, and then disabled right when I_L reaches zero. Therefore, in either example, it is necessary to monitor I_L to accurately determine when I_L reaches zero. In one embodiment, ZC comparator **114** is coupled to node **124** in order to monitor I_L and assert ZC when I_L reaches zero. ZC can also be provided to control circuitry **126** which disables (i.e. turns off) transistor **104** in response to assertion of ZC (i.e. in response to ZC indicating a zero crossing condition for I_L). Transistor **104** can therefore be maintained off until the next on-state. ZC comparator **114** can also be implemented using the current comparator which will be described below in reference to FIG. 2.

[0016] FIG. 2 illustrates, in partial block diagram and partial schematic form, a current comparator **200** which can be used to implement each of ZC comparator **114** and OC comparator **116**, in accordance with one embodiment of the present invention. Comparator **200** includes a first input V_{ip} and a second input V_{in} , in which V_{ip} and V_{in} are coupled as inputs to a differential stage of comparator **200** (corresponding to Stage 1 of FIG. 2). The differential stage of comparators **200** includes an input capacitor **C0** **202** (which may simply be referred to as **C0** herein) having a first terminal coupled to receive V_{ip} and a second input capacitor **C1** **204** (which may simply be referred to as **C1** herein) having a first terminal coupled to receive V_{in} . The differential stage also includes a differential amplifier **260** in which a second terminal of **C0** is coupled to a non-inverting (+) input of differential amplifier **260** and a second terminal of **C1** is coupled to an inverting (−) input of differential amplifier **260**. Differential amplifier **260** includes a current source **206**, PMOS transistors **P0** **208** and **P1** **210** and NMOS transistors **N0** **212** and **N1** **214**. Note that these PMOS and NMOS transistors may simply be referred to as **P0**, **P1**, **N0**, and **N1**, respectively. Differential amplifier **260** (implemented in the illustrated embodiment as a common-source amplifier) also includes NMOS transistors **216** and **218**, each having a control electrode coupled to receive an auto-zero (AZ) control signal (which may simply be referred to as **AZ**). Comparator **200** also includes a cascode amplifier **262** (corresponding to Stage 2 of FIG. 2) coupled to an output of differential amplifier **260**, and an inverter **234** having an input coupled to an output of cascode amplifier **262** and an output coupled to provide an output V_o . In the illustrated embodiment, V_o is provided by inverter **234** as a digital logic signal.

[0017] In differential amplifier **260**, a current source **206** is coupled between V_{dd} and circuit node **236**. A first current electrode (e.g. source) of **P0** is coupled to circuit node **236**, a control electrode of **P0** is coupled to the second terminal of **C0**, and a second current electrode (e.g. drain) of **P0** is coupled to a circuit node **238**. A first current electrode (e.g. source) of **P1** is coupled to node **236**, a control electrode of **P1** is coupled to the second terminal of **C1**, and a second current electrode (e.g. drain) of **P1** is coupled to a circuit node **242**. A first current electrode (e.g. drain) of **N0** and a control electrode of **N0** are coupled to node **238**, a second current electrode (e.g. source) of **N0** is coupled to V_{ss} . A first current electrode (e.g. drain) of **N1** is coupled to node **242**, a control electrode of **N1** is coupled to the control electrode of **N0**, and a second current electrode (e.g. source) of **N1** is coupled to V_{ss} .

[0018] Control switches (e.g. transistors **216** and **218**) of differential amplifier **260** are coupled between the gate and drains of **P0** and **P1** to selectively short the gate to drains based on the value of **AZ**. Transistor **216** has a first current electrode coupled to the control electrode of **P0** at a circuit node **240**, a second current electrode coupled to the second current electrode (e.g. drain) of **P0** at node **238**, and a control electrode coupled to receive **AZ**. Transistor **218** has a first current electrode coupled to the control electrode of **P1** at a circuit node **244**, a second current electrode coupled to the second current electrode of **P1** at node **242**, and a control electrode coupled to receive **AZ**. Therefore, when **AZ** is asserted to a logic level one, the gates of **P0** and **P1** are each shorted to their corresponding drains. When **AZ** is negated to a logic level 0, transistors **216** and **218** are off, in which the gates of **P0** and **P1** are no longer shorted to their corresponding drains.

[0019] Cascode amplifier **262** includes a current source **232**, capacitors **230** and **220**, and NMOS transistors **N2** **222** and **N3** **224** (which may simply be referred to as **N2** and **N3**, respectively). Cascode amplifier **262** also includes NMOS transistors **226** and **228**, each having a control electrode coupled to receive **AZ**. Current source **232** is coupled between **Vdd** and a circuit node **250**, which corresponds to an output of cascode amplifier **262**. A first current terminal (e.g. drain) of **N3** is coupled to node **250**, a control electrode of **N3** is coupled to a first terminal of a capacitor **230** at a circuit node **246**, and a second current electrode of **N3** is coupled to a circuit node **248**. A capacitor **230** has a first terminal coupled to node **246** and a second terminal coupled to **Vss**. A first current electrode (e.g. drain) of **N2** is coupled to the second current electrode of **N3** at node **248**, a second current electrode (e.g. source) of **N2** is coupled to **Vss**. A first terminal of capacitor **220** is coupled to node **242** (corresponding to an output of differential amplifier **260**), and a second terminal of capacitor **220** is coupled to a control electrode of **N2**.

[0020] Control switches (e.g. transistors **226** and **228**) of cascode amplifier **262** are coupled between the gate and drains of **N3** and **N2** to selectively short the gate to drains based on the value of **AZ**. Transistor **226** has a first current electrode coupled to the control electrode of **N3** at circuit node **246**, a second current electrode coupled to the first current electrode (e.g. drain) of **N3** at node **250**, and a control electrode coupled to receive **AZ**. Transistor **228** has a first current electrode coupled to the control electrode of **N2** at circuit node **252**, a second current electrode coupled to the first current electrode of **N2** at node **248**, and a control electrode coupled to receive **AZ**. Therefore, when **AZ** is asserted to a logic level one, the gates of **N3** and **N2** are each shorted to their corresponding drains. When **AZ** is negated to a logic level 0, transistors **226** and **230** are off, in which the gates of **N3** and **N2** are no longer shorted to their corresponding drains.

[0021] An inverter **234** has an input coupled to node **250**, corresponding to the output of cascode amplifier **262**. Inverter **234** has an output coupled to provide **Vo**. Note that, although illustrated as a single inverter, inverter **234** can be implemented with a series of inverters (such as a series of an odd number of inverters) to apply more gain and further digitize (e.g. sharpen) the signal at **Vo**. Note that inverter **234** can also be referred to as a logic stage.

[0022] In operation, control circuitry **126** controls **AZ** so as to operate comparator **200** of FIG. 2 in two stages: an

auto-zero phase (in which **AZ** is asserted to a logic level one) and a compare phase (in which **AZ** is negated to a logic level zero). In one embodiment, as will be described in reference to FIGS. 3-5 below, comparator **200** can be used in ZC comparator **114** in which **Vo** (corresponding to ZC) is asserted when I_L has fallen to zero. In another embodiment, as will be described in reference to FIG. 6 below, comparator **200** can be used in OC comparator **116** in which **Vo** (corresponding to OC) is asserted when I_L has reached a maximum current, I_{max} (which can be provided by a replica circuit **118**, also illustrated in FIG. 6).

[0023] FIG. 3 illustrates, in partial schematic and partial block diagram form, a more detailed view of ZC comparator **114** which includes a front-end input signal switch **300** and a comparator, such as comparator **200** of FIG. 2, which includes input capacitors **C0** **302** and **C1** **304** (which are analogous to **C0** and **C1**, respectively, of comparator **200**), a Stage 1 with a differential amplifier **360** (which may be implemented with differential amplifier **260** of FIG. 2), a Stage 2 with a cascode amplifier **362** (which may be implemented with differential amplifier **262** of FIG. 2), and an inverter **334** (which is analogous to inverter **234** of FIG. 2). Therefore, the descriptions provided for comparator **200** of FIG. 2 apply to the comparator of FIG. 3. Input signal switch **300** is coupled to provide the appropriate inputs to **Vip** and **Vin**, respectively, for the auto-zero phase and compare phase. When **AZ** is asserted to a logic level one, indicating the auto-zero phase, switches **308** and **306** of input switch **300** are set to their first position to provide **Vip_az** to **Vip** and **Vin_az** to **Vin**, and when **AZ** is asserted to a logic level zero, indicating the compare phase, switches **308** and **306** of input switch **300** are set to their second position to provide **Vip_cmp** to **Vip** and **Vin_cmp** to **Vin**. For ZC comparator **114**, each of **Vip_az**, **Vin_az**, and **Vin_cmp** are coupled to **Vss**, and **Vip_cmp** is coupled to **Vx** (at node **124** of converter **100** of FIG. 1).

[0024] In operation, during the **AZ** phase of ZC comparator **114**, switches **308** and **306** are set in their first position in which both **Vin** and **Vip** are coupled to **Vss** (which may be provided by a local AC ground). Note that **Vo** of FIG. 3, when operated with the two phases, can be expressed as " $Vo = A[(Vip_cmp - Vin_cmp) - (Vip_az - Vin_az)]$ " in which **A** refers to a gain of the comparator and is generally large enough that **Vo** will saturate at either its positive or negative supply. In the case of ZC comparator **114**, during the auto-zero phase, **Vip_az**=**Vin_az**, which causes these terms to drop out of the equation, in addition to any offsets naturally present in the circuit. Operation of ZC comparator **114** will be described in reference to the elements of comparator **200** of FIG. 2.

[0025] During the auto-zero phase (with **AZ**=1), switches **216**, **218**, **226**, and **230** (also referred to as auto-zero switches) are all turned on, shorting the corresponding gates and drains. The resulting circuit, using the elements of comparator **200** of FIG. 2, is illustrated in FIG. 4. As illustrated in FIG. 4, the gate of **P0** is shorted to node **238** and the gate of **P1** is shorted to node **242**. With **Vin** and **Vip** each coupled to ground, the voltage over **C0** (**V(C0)**) can be expressed as $V(C0) = V_{TN0}$, in which V_{TN0} is the threshold voltage (**Vt**) of **N0** (i.e. the voltage at node **238**). Similarly, the voltage over **C1** (**V(C1)**) can be expressed as $V(C1) = V_{TN0} + V_{TP0} - V_{TP1}$, in which V_{TP0} and V_{TP1} correspond to the **Vt** of each of **P0** and **P1**, respectively. The offset between **C0** and **C1** can be expressed as $V(C0) - V(C1) = V_{TP1} - V_{TP0}$.

(since the V_{TN0} 's cancel each other) and the amplifier's natural offset is $V_{TP0}-V_{TP1}$. In cascode amplifier 262, as illustrated in FIG. 4, the gate of N3 is shorted node 250, and the gate of N2 is shorted to node 248.

[0026] The offset between C0 and C1 is stored on capacitors C0 and C1 at the input of Stage 1, and capacitor 220 serves as a DC block to keep stage 2 biased in a high-gain state. Capacitor 230 also serves a similar purpose in maintaining the biasing condition established during the auto-zero phase. Note that during the auto-zero phase, the voltage at node 250 can be expressed as " $V_{TN2}+V_{TN3}$ " (in which V_{TN2} and V_{TN3} correspond to the V_t 's of N2 and N3, respectively). This voltage at node 250 (which is about two V_t 's) is near the switching threshold of inverter 234. That is, the capacitors of comparator 200 operate to store a voltage at the input of inverter 234 which is close to its switching threshold. This allows for improved precision when receiving smaller input voltages, or input voltages having a smaller voltage difference between them, by cancelling any offsets in the amplifier.

[0027] During the compare phase (with $AZ=0$), switches 216, 218, 226, and 230 are turned off, thus placed into a high-Z state. The resulting circuit, using the elements of comparator 200 of FIG. 2, is illustrated in FIG. 5. During the compare phase, Stage 1 behaves as a differential amplifier and Stage 2 as a single-ended cascode amplifier. Referring to FIG. 2 (in which switches 308 and 306 are set to their second position for the compare phase), the "+" input to differential amplifier 360 is labeled as V_{ip_ac} in which $V_{ip_ac}=V_x+V(C0)$, and the "-" input is labeled as V_{in_ac} in which $V_{in_ac}=0+V(C1)$. In this case, $V_{ip_ac}-V_{in_ac}=V_x+V(C0)-V(C1)$. Therefore, it can be seen that the offset in the compare phase ($V(C0)-V(C1)$) is canceled by the offset stored in the auto-zero phase. Also, with the input to inverter 234 being set close to its switching threshold in the auto-zero phase, the zero crossing (ZC) by V_x can be quickly and accurately detected (resulting in asserting ZC to a logic level one).

[0028] As indicated above, a comparator such as comparator 200 can also be used in OC comparator 116. Therefore, FIG. 6 illustrates, in partial schematic and partial block diagram form, a more detailed view of OC comparator 116 which includes a front-end input signal switch 600 (analogous to input signal switch 300) and a comparator, such as comparator 200 of FIG. 2, which includes input capacitors C0 602 and C1 604 (which are analogous to C0 and C1, respectively, of comparator 200), a Stage 1 with a differential amplifier 660 (which may be implemented with differential amplifier 260 of FIG. 2), a Stage 2 with a cascode amplifier 662 (which may be implemented with differential amplifier 262 of FIG. 2), and an inverter 634 (which is analogous to inverter 234 of FIG. 2). Therefore, the descriptions provided for comparator 200 of FIG. 2 also apply to the comparator of FIG. 6. Input signal switch 600 is coupled to provide the appropriate inputs to V_{ip} and V_{in} , respectively, for the auto-zero phase and compare phase. When AZ is asserted to a logic level one, indicating the auto-zero phase, switches 608 and 606 of input switch 600 are set to their first position to provide V_{ip_az} to V_{ip} and V_{in_az} to V_{in} , and when AZ is asserted to a logic level zero, indicating the compare phase, switches 608 and 606 of input switch 600 are set to their second position to provide V_{ip_cmp} to V_{ip} and V_{in_cmp} to V_{in} . For OC comparator 116, each of V_{ip_az} and V_{in_az} are coupled to Vdd, V_{in_cmp} is coupled

to receive $VR_{I_{max}}$ (which corresponds to a voltage input from a reference circuit which generates the maximum current, I_{max} , for the current rating of inductor 110), and V_{in_cmp} is coupled to V_x (node 124 of converter 100 of FIG. 1).

[0029] Referring back to FIG. 1, note that replica circuit 118 (intended to replicate transistor 102 and V_x) includes a PMOS transistor 120 having a first current electrode coupled to Vdd, a control electrode coupled to Vss, and a second current electrode coupled to a first terminal of a current source 122. A second terminal of current source 122 is coupled to Vss. Current source 122 may be set, for example, to I_{max}/M in which M represents the sizing of transistor 102 with respect to transistor 120. The second current electrode of transistor 120 provides the voltage $VR_{I_{max}}$ to OC comparator 116. In the illustrated embodiment, note that, because transistor 120 is a scaled replica of transistor 102, when I_L through inductor 110 reaches I_{max} , the voltage at $VR_{I_{max}}$ will equal V_x , therefore, this comparison can be used to detect an OC condition (similar to how the comparison between zero and V_x is used to detect a ZC condition). Note that alternate embodiments may use other types of circuitry, such as other replica circuits, to provide an indication of I_{max} to OC comparator 116.

[0030] Therefore, during the auto-zero phase of OC comparator 116, rather than setting both V_{in} and V_{ip} to Vss, both are set to Vdd. However, since $V_{ip_az}=V_{in_az}$ in this case as well, these terms also drop out of the equation. With $AZ=1$ during the auto-zero phase (in which switches 606 and 608 are in their first position), comparator 200 is configured as was described above with respect to FIG. 4 (in which the auto-zero switches are all turned on), which results in again storing the offset $V(C0)-V(C1)$ at the input capacitors C1 and C0. With $AZ=0$ during the compare phase (in which switches 606 and 608 are in their second position), comparator 200 is configured as was described above with respect to FIG. 5 (in which the auto-zero switches are off). In this case, though, rather than compare V_x ($V_{ip}=V_x$) to 0 ($V_{in}=0$) to determine a ZC condition, $VR_{I_{max}}$ ($V_{ip}=VR_{I_{max}}$) is compared to V_x ($V_{in}=V_x$) to determine an OC condition.

[0031] Note that the descriptions provided above with respect to FIGS. 4 and 5 for the operation of comparator 200 when used in ZC comparator 114 are similar to the operation of comparator 200 when used in OC comparator 116. Converter 100 may include either ZC comparator 114 or OC comparator 116 or may include both (as in the illustrated embodiment of FIG. 1). In an embodiment in which converter 100 includes both the ZC and OC comparators, two different instantiations of current comparator 200 are required, one for the ZC comparator and one for the OC comparator. Each of the outputs ZC and OC are provided to control circuitry 126 so that control circuitry 126 can disable transistor 104 in response to a ZC condition occurring (i.e. in response to assertion of ZC) and can disable transistor 102 in response to an OC condition occurring (i.e. in response to assertion of OC). Note that comparator 200 can be used in other embodiments as well, such as in different types of DC-DC converters.

[0032] With the use of comparator 200 of FIG. 2, note that the Miller effect may be mitigated or defeated. As known in the art, the Miller effect accounts for an increase in the equivalent input capacitance of an inverting voltage amplifier due to amplification of the effect of capacitance between the input and output terminals. For example, for an inverting

amplifier $V_{out} = -A_v \cdot V_{in}$ in which $-A_v$ is the voltage gain of the inverting amplifier and a feedback capacitance (C) is present between V_{in} and V_{out} . However, the increased input capacitance due to the Miller effect can be expressed as $C_M = C(1 + A_v)$. With the use of comparator **200**, however, the large virtual capacitance which would appear at the input to differential amplifier **260** due to the Miller effect can be mitigated or overcome. For example, since the inputs of the first stage (V_{in} and V_{ip}) are coupled to low impedance power switches (e.g. transistors **102** and **104**), a faster RC time constant is achievable thus reducing the impact of C_M . Also, by having a lower voltage gain, the effect from C_M is smaller. In comparator **200**, cascode amplifier **262** of Stage **2** helps achieve a lower voltage gain from node **252** to node **248** for the comparator, thereby mitigating the Miller effect.

[0033] For example, referring to comparator **200** of FIG. **1**, assuming that the offset is fully cancelled at input capacitors **202** and **204**, if ($V_{in} > V_{ip}$) is initially applied, the output of Stage **1** (at node **242**) will initially be low, the output of Stage **2** will initially be high, and V_o will initially be low. In this configuration, the transistors of Stage **2** are kept in saturation, and thus amplifier **262** behaves as a cascoded amplifier (reducing the Miller effect) and current for the falling edge is set by the common-source transistor's transconductance. On the other hand, if ($V_{ip} > V_{in}$) is initially applied, the output of Stage **2** (at node **250**) is low, the pull-up current is limited by the bias current and amplifier transistors are in triode mode, which increases their channel capacitance, so the circuit will respond significantly more slowly.

[0034] Unlike other auto-zeroed comparators in analog-to-digital converters (ADCs), auto-zeroed comparators in DC-DC converters for current sensing thresholds will naturally have their inputs slew in one specification direction each cycle. For example, in a high-side PMOS power switch in a buck converter (e.g. transistor **102**), the inductor current I_L will increase when the high-side is enabled, and the drain voltage (e.g. V_x) will become progressively lower than the supply. Similarly, with a low-side NMOS power switch in a buck converter (e.g. transistor **104**), the drain voltage (e.g. V_x) will consistently become more positive. Because the slew direction of the input voltage is pre-determined, the converter circuit can be optimized for response time to a given input slew.

[0035] Also, note that comparator **200** performs an earlier conversion from differential to single-ended signals (through differential amplifier **260**, prior to Stage **2**). This permits reducing the bias current of some stages (e.g. in half) when compared to other comparator designs in which more stages are each kept fully differential. Note also that at least having Stage **1** receive differential inputs allows for lower voltage converters as opposed to using a single ended Stage **1** due to problems for a single ended first stage caused by variations in the ground potential.

[0036] In alternate embodiments, comparator **200** may instead use an NMOS differential pair as opposed to **P0** and **P1**. Also, additional gain stages may be used in addition to those illustrated (e.g. in addition to cascode amplifier **262**). However, if so, they should alternate between NMOS and PMOS cascode amplifiers to achieve the intended high-speed operation. Also, in alternate embodiments, the auto-zeroing switches (e.g. switches **216**, **218**, **226**, and **230**) can instead be implemented as PMOS switches or may be implemented using a different combination of circuitry.

Alternate embodiments may also use different power switch topologies than single transistor switches illustrated for switches **104** and **102**, and the corresponding simple replica switch **120**. For example, FIG. **7** illustrates in an embodiment of the power stage of converter **100** which implements the low-side switch as two NMOS transistors in series (e.g. transistor **104** in series with an NMOS transistor **702**) between PMOS transistor **102** and the second voltage supply terminal. In this example, front-end input signal switches **300** and **600** may instead be coupled to a second internal node **704** of the power stage located between these two series-connected NMOS transistors **104** and **702** for use during the compare phases of ZC **114** or OC **116**, rather than using the first internal node, V_x , directly. Also note that different switch topologies may be used to implement the functionality of the front-end input signal switches (e.g. switches **300** and **600**). For example, multiple switches or devices may be used to implement each of the input switches (e.g. **308**, **306**, **608**, **606**). Also, while inverter **234** forms an output inverter for comparator **200**, as described above, inverter **234** (also referred to as logic stage **234**) can be implemented in a variety of different ways, such as including more than one inverter or can use a more complicated logic circuit (such as with NAND or NOR gates, or other combinations of gates, to block leakage current during the auto-zeroing phase or to give a defined state when disabled).

[0037] Note that DC-DC comparator **100** may be implemented using split grounds in which different voltage supply terminals (e.g. ground terminals) may be used for different portions of DC-DC comparator **100** to provide ground voltages (in which, in one embodiment, the ground terminals can be connected or shorted to each other at the printed circuit board (PCB)). In one embodiment, the second voltage supply terminal (e.g. ground terminal) which is configured to provide V_{ss} (e.g. a ground signal) for the power stage may be a different ground terminal as used for ZC comparator **114** or OC comparator **116**. For example, the ground terminal for the comparators may be a third voltage supply terminal configured to provide a second ground signal, but this second ground signal may be different from the ground signal provided by the second voltage supply terminal to the power stage. That is, the supplied V_{ss} voltages may be different between the power stage and comparators (but are all still less than V_{dd}). Furthermore, based on the embodiment, different ground signals may be used for the AZ phase vs the compare phase for each comparator. For example, in ZC comparator **114**, front-end switch **300** may connect the third voltage supply terminal to the V_{ip} and V_{in} inputs of stage **1**, but connect the second voltage supply terminal to the V_{in} input for the compare phase. Similarly, note that different voltage supply terminals may be used for different portions of DC-DC comparator **100** to provide supply voltages. For example, a fourth voltage supply terminal may be used to provide V_{dd} to ZC comparator **114** or OC comparator **116**, in which this V_{dd} may be lower than the V_{dd} provided by the first voltage supply terminal to the power stage. For example, the V_{dd} for the power stage may be 3.3V while the V_{dd} for the comparators may be 1.8V. This may help protect lower voltage devices within the comparators. Note also that for OC comparator **116**, since it is preferable that the V_{dds} used as inputs during the AZ phase are similar to the V_{dd} used during the compare phase, even if the V_{dd} for the comparators is lower as compared to the power stage (e.g. 1.8V instead of 3.3V), the V_{dds} used as

inputs during the AZ phase as well as the V_{dd} used during the comparison phase may all be the V_{dd} which is applied to the power stage (e.g. the higher 3.3V).

[0038] Therefore, by now it can be appreciated that a current comparator configuration has been provided which can be used in various threshold current comparators, such as in an ZC comparator or an OC comparator. In one embodiment, the current comparator includes a first stage with a differential amplifier and a second stage with a cascode amplifier, and the current comparator operates in an auto-zero phase followed by a compare phase. In the auto-zero phase, auto-zeroing switches are used to tie gates and drains of various transistors in each of the first and second stages in order to store an offset into input capacitors during an auto-zeroing phase. During the compare phase, the auto-zeroing switches are turned off so as to allow the differential amplifier and cascode amplifiers to operate normally. The initial stored offset cancels the offset during the compare phase in which a first input voltage is compared with a second input voltage. An output of the comparator indicates when the first input voltage has reached the second input voltage. Such a current comparator may be used as the ZC comparator or OC comparator of, for example, a DC-DC buck converter. For example, for the ZC comparator within a buck converter, the output of the current comparator indicates when V_x at the first input voltage (representative of the current through the inductor of the buck converter) has reached 0 at the second input voltage. For an OC comparator within the buck converter, the output of the current comparator indicates when a V_{RI_{max}} at the first input voltage (the voltage representative of a maximum current allowed through the inductor) has reached V_x at the second input voltage.

[0039] The terms “assert” or “set” and “negate” (or “deassert” or “clear”) are used herein when referring to the rendering of a signal, status bit, or similar apparatus into its logically true or logically false state, respectively. If the logically true state is a logic level one, the logically false state is a logic level zero. And if the logically true state is a logic level zero, the logically false state is a logic level one.

[0040] Each signal described herein may be designed as positive or negative logic, where negative logic can be indicated by a bar over the signal name or an asterisk (*) following the name. In the case of a negative logic signal, the signal is active low where the logically true state corresponds to a logic level zero. In the case of a positive logic signal, the signal is active high where the logically true state corresponds to a logic level one. Note that any of the signals described herein can be designed as either negative or positive logic signals. Therefore, in alternate embodiments, those signals described as positive logic signals may be implemented as negative logic signals, and those signals described as negative logic signals may be implemented as positive logic signals.

[0041] Because the apparatus implementing the present invention is, for the most part, composed of electronic components and circuits known to those skilled in the art, circuit details will not be explained in any greater extent than that considered necessary as illustrated above, for the understanding and appreciation of the underlying concepts of the present invention and in order not to obfuscate or distract from the teachings of the present invention.

[0042] Although the invention has been described with respect to specific conductivity types or polarity of poten-

tials, skilled artisans appreciated that conductivity types and polarities of potentials may be reversed.

[0043] Some of the above embodiments, as applicable, may be implemented using a variety of different applications. For example, although FIG. 1 and the discussion thereof describe an exemplary configuration of a DC-DC converter, this exemplary configuration is presented merely to provide a useful reference in discussing various aspects of the invention. Therefore, the description of the architecture has been simplified for purposes of discussion, and it is just one of many different types of converters and configurations that may be used in accordance with the invention. Those skilled in the art will recognize that the boundaries between logic blocks are merely illustrative and that alternative embodiments may merge logic blocks or circuit elements or impose an alternate decomposition of functionality upon various logic blocks or circuit elements. Thus, it is to be understood that the configurations depicted herein are merely exemplary, and that in fact many other configurations can be implemented which achieve the same functionality.

[0044] Also for example, in one embodiment, the illustrated elements of converter 10 are circuitry located on a single integrated circuit or within a same device, in which, for example, converter 10 can be implemented in a system on a chip (SoC). Alternatively, converter 10 may be used in a system which includes any number of separate integrated circuits or separate devices interconnected with each other.

[0045] Furthermore, those skilled in the art will recognize that boundaries between the functionality of the above described operations merely illustrative. The functionality of multiple operations may be combined into a single operation, and/or the functionality of a single operation may be distributed in additional operations. Moreover, alternative embodiments may include multiple instances of a particular operation, and the order of operations may be altered in various other embodiments.

[0046] Although the invention is described herein with reference to specific embodiments, various modifications and changes can be made without departing from the scope of the present invention as set forth in the claims below. For example, the current comparator can be used in other applications, such as in other types of threshold comparators or in other types of DC-DC converters. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present invention. Any benefits, advantages, or solutions to problems that are described herein with regard to specific embodiments are not intended to be construed as a critical, required, or essential feature or element of any or all the claims.

[0047] The term “coupled,” as used herein, is not intended to be limited to a direct coupling or a mechanical coupling.

[0048] Furthermore, the terms “a” or “an,” as used herein, are defined as one or more than one. Also, the use of introductory phrases such as “at least one” and “one or more” in the claims should not be construed to imply that the introduction of another claim element by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim element to inventions containing only one such element, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an.” The same holds true for the use of definite articles.

[0049] Unless stated otherwise, terms such as “first” and “second” are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements.

[0050] The following are various embodiments of the present invention. Note that any of the aspects below can be used in any combination with each other and with any of the disclosed embodiments.

[0051] In one embodiment, a current comparator includes a first capacitor having a first terminal coupled to a first input signal; a second capacitor having a first terminal coupled to a second input signal; a first transistor having a first current electrode coupled to a first voltage supply terminal via a first current source, a control electrode coupled to a second terminal of the first capacitor, and a second current electrode coupled to a first circuit node; a second transistor having a first current electrode coupled to the first current electrode of the first transistor, a control electrode coupled to a second terminal of the second capacitor, and a second current electrode coupled to a second circuit node; a third transistor having a first current electrode coupled to the first circuit node, a second current electrode coupled to a second voltage supply terminal, and a control electrode coupled to the first circuit node; a fourth transistor having a first current electrode coupled to the second circuit node, a control electrode coupled to the control electrode of the third transistor, and a second current electrode coupled to the second voltage supply terminal; a single-ended cascode amplifier having an input coupled via a third capacitor to the second circuit node and an output; a logic stage coupled to receive the output of the single-ended cascode amplifier, and configured to provide an output of the current comparator; and a set of auto-zero switches configured to, in response to an auto-zero control signal, selectively short the control electrode of the first transistor to the second current electrode of the first transistor and selectively short the control electrode of the second transistor to the second current electrode of the second transistor. In one aspect, the single-ended cascode amplifier includes a fifth transistor having a control electrode configured as the input of the single-ended cascode amplifier, a first current electrode coupled to the second voltage supply terminal, and a second current electrode; and a sixth transistor having a first current electrode coupled to the second current electrode of the fifth transistor, a control electrode, and a second current electrode coupled to the output of the single-ended cascode amplifier, wherein the set of auto-zero switches is further configured to, in response to the auto-zero control signal, selectively short the control electrode of the fifth transistor to the second current electrode of the fifth transistor and selectively short the control electrode of the sixth transistor to the second current electrode of the sixth transistor. In a further aspect, the control electrode of the sixth transistor is coupled to the second voltage supply terminal via a fourth capacitor. In another further aspect, the current comparator is configured to operate in an auto-zero phase followed by a compare phase, wherein, during the auto-phase, the set of auto-zero switches are configured to short the control electrode to the second current electrode of each of the first, second, fifth, and sixth transistors, and during the compare-phase, the set of auto-zero switches is each in a high-impedance state. In yet a further aspect, each of the first and second transistors are PMOS transistors and each of the third, fourth, fifth, and

sixth transistors are NMOS transistors. In yet an even further aspect, each of the set of auto-zero switches is implemented as an NMOS transistor. In another aspect of the embodiment, the current comparator is configured to operate in an auto-zero phase followed by a compare phase, wherein, during the auto-phase, the set of auto-zero switches are configured to short the control electrode to the second current electrode of each of the first and second transistors, and during the compare-phase, the set of auto-zero switches is each in a high-impedance state.

[0052] In another aspect of the embodiment, a DC-DC converter includes the current comparator of the embodiment, and an input signal switch configured to provide auto-zero phase inputs as the first and second input signals of the current comparator during the auto-zero phase and to provide compare phase inputs as the first and second input signals of the current capacitor during the compare phase. In a further aspect of the DC-DC converter, the auto-zero phase inputs provided as the first and second input signals during the auto-zero phase are a same signal, and the compare phase inputs provided as the first and second input signals during the compare phase are a pair of signals being compared such that the output of the current comparator is provided based on the comparison of the pair of signals. In another aspect of the DC-DC converter, the DC-DC converter further includes a power switch having a PMOS transistor in series with an NMOS transistor, wherein a first internal circuit node between the PMOS transistor and the NMOS transistor is configured to be coupled to an inductor. In a further aspect of the DC-DC converter, during the auto-zero phase, the input signal switch couples a first ground signal to each of the first and second input signals of the current comparator, and, during the compare phase, the input signal switch couples a second internal circuit node of the power switch and a second ground signal to the first and second input signals of the current comparator, wherein the output of the current comparator indicates when a voltage on the second internal circuit node reaches a voltage of the second ground signal. In yet a further aspect of the DC-DC converter, the DC-DC converter further includes control circuitry, wherein when the output of the current comparator is asserted to indicate that the voltage on the second internal circuit node has reached the voltage of the second ground signal, the control circuitry disables the NMOS transistor of the power switch. In yet a further aspect, the power switch further comprises an additional NMOS transistor coupled between the NMOS transistor of the power switch and a third voltage supply terminal configured to provide the second ground signal, the second internal circuit node of the power switch corresponds to a node between the NMOS transistor and additional NMOS transistor of the power switch, and the first ground signal is provided by one of the second voltage supply terminal or the third voltage supply terminal. In another aspect of the DC-DC converter, during the auto-zero phase, the input signal switch couples a supply voltage to each of the first and second input signals of the current comparator, and, during the compare phase, the input signal switch couples a voltage indicative of a predetermined maximum current through the inductor and the internal circuit node of the power switch to the first and second input signals of the current comparator, wherein the output of the current comparator indicates when a voltage on the internal circuit node reaches the voltage indicative of the predetermined maximum current. In a further aspect, the

DC-DC further includes a replica circuit configured to provide the voltage indicative of the predetermined maximum current to the input signal switch.

[0053] In another embodiment, a current comparator includes a first capacitor having a first terminal coupled to a first input signal; a second capacitor having a first terminal coupled to a second input signal; a first transistor having a first current electrode coupled to a first voltage supply terminal via a first current source, a control electrode coupled to a second terminal of the first capacitor, and a second current electrode coupled to a first circuit node; a second transistor having a first current electrode coupled to the first current electrode of the first transistor, a control electrode coupled to a second terminal of the second capacitor, and a second current electrode coupled to a second circuit node; a third transistor having a first current electrode coupled to the first circuit node, a second current electrode coupled to a second voltage supply terminal, and a control electrode coupled to the first circuit node; a fourth transistor having a first current electrode coupled to the second circuit node, a control electrode coupled to the control electrode of the third transistor, and a second current electrode coupled to the second voltage supply terminal; and a single-ended cascode amplifier having an input coupled via a third capacitor to the second circuit node and an output, wherein the current comparator is configured to operate in an auto-zero phase followed by a compare phase, wherein, during the auto-phase, the current comparator is configured to short the control electrode to the second current electrode of each of the first and second transistors, and during the compare phase, the current comparator is configured to not short the control electrode to the second current electrode of each of the first and second transistors. In one aspect of the another embodiment, the current comparator further includes a logic stage coupled to receive the output of the single-ended cascode amplifier, and configured to provide an output of the current comparator. In another aspect, the single-ended cascode amplifier includes a fifth transistor having a control electrode configured as the input of the single-ended cascode amplifier, a first current electrode coupled to the second voltage supply terminal, and a second current electrode; and a sixth transistor having a first current electrode coupled to the second current electrode of the fifth transistor, a control electrode, and a second current electrode coupled to the output of the single-ended cascode amplifier, wherein the current comparator is further configured to, during the auto-zero phase, short the control electrode of the fifth transistor to the second current electrode of the fifth transistor and short the control electrode of the sixth transistor to the second current electrode of the sixth transistor, and, during the compare phase, not short the control electrode to the second current electrode of each of the fifth and sixth transistors.

[0054] In another aspect, a DC-DC converter includes the current comparator of the another embodiment, and a power switch having a PMOS transistor in series with an NMOS transistor, wherein an internal circuit node between the PMOS transistor and the NMOS transistor is configured to be coupled to an inductor, and wherein, during the auto-zero phase, a first ground signal is provided as each of the first and second input signals of the current comparator, and, during the compare phase, a second internal circuit node of the power switch and a second ground signal are provided to the first and second input signals of the current comparator,

wherein the output of the current comparator indicates when a voltage on the second internal circuit node reaches a voltage of the second ground signal.

[0055] In another aspect, a DC-DC converter includes the current comparator of the another embodiment, and a power switch having a PMOS transistor in series with an NMOS transistor, wherein an internal circuit node between the PMOS transistor and the NMOS transistor is configured to be coupled to an inductor, and wherein, during the auto-zero phase, a supply voltage is provided as each of the first and second input signals of the current comparator, and, during the compare phase, a voltage indicative of a predetermined maximum current through the inductor and the internal circuit node are coupled to the first and second input signals of the current comparator, wherein the output of the current comparator indicates when a voltage on the internal circuit node reaches the voltage indicative of the predetermined maximum current.

What is claimed is:

1. A current comparator, comprising:

- a first capacitor having a first terminal coupled to a first input signal;
- a second capacitor having a first terminal coupled to a second input signal;
- a first transistor having a first current electrode coupled to a first voltage supply terminal via a first current source, a control electrode coupled to a second terminal of the first capacitor, and a second current electrode coupled to a first circuit node;
- a second transistor having a first current electrode coupled to the first current electrode of the first transistor, a control electrode coupled to a second terminal of the second capacitor, and a second current electrode coupled to a second circuit node;
- a third transistor having a first current electrode coupled to the first circuit node, a second current electrode coupled to a second voltage supply terminal, and a control electrode coupled to the first circuit node;
- a fourth transistor having a first current electrode coupled to the second circuit node, a control electrode coupled to the control electrode of the third transistor, and a second current electrode coupled to the second voltage supply terminal;
- a single-ended cascode amplifier having an input coupled via a third capacitor to the second circuit node and an output;
- a logic stage coupled to receive the output of the single-ended cascode amplifier, and configured to provide an output of the current comparator; and
- a set of auto-zero switches configured to, in response to an auto-zero control signal, selectively short the control electrode of the first transistor to the second current electrode of the first transistor and selectively short the control electrode of the second transistor to the second current electrode of the second transistor.

2. The current comparator of claim 1, wherein the single-ended cascode amplifier comprises:

- a fifth transistor having a control electrode configured as the input of the single-ended cascode amplifier, a first current electrode coupled to the second voltage supply terminal, and a second current electrode; and
- a sixth transistor having a first current electrode coupled to the second current electrode of the fifth transistor, a control electrode, and a second current electrode

coupled to the output of the single-ended cascode amplifier, wherein the set of auto-zero switches is further configured to, in response to the auto-zero control signal, selectively short the control electrode of the fifth transistor to the second current electrode of the fifth transistor and selectively short the control electrode of the sixth transistor to the second current electrode of the sixth transistor.

3. The current comparator of claim 2, wherein the control electrode of the sixth transistor is coupled to the second voltage supply terminal via a fourth capacitor.

4. The current comparator of claim 2, wherein the current comparator is configured to operate in an auto-zero phase followed by a compare phase, wherein, during the auto-phase, the set of auto-zero switches are configured to short the control electrode to the second current electrode of each of the first, second, fifth, and sixth transistors, and during the compare-phase, the set of auto-zero switches is each in a high-impedance state.

5. The current comparator of claim 4, wherein each of the first and second transistors are PMOS transistors and each of the third, fourth, fifth, and sixth transistors are NMOS transistors.

6. The current comparator of claim 5, wherein each of the set of auto-zero switches is implemented as an NMOS transistor.

7. The current comparator of claim 1, wherein the current comparator is configured to operate in an auto-zero phase followed by a compare phase, wherein, during the auto-phase, the set of auto-zero switches are configured to short the control electrode to the second current electrode of each of the first and second transistors, and during the compare-phase, the set of auto-zero switches is each in a high-impedance state.

8. A DC-DC converter comprising:

the current comparator of claim 1;

an input signal switch configured to provide auto-zero phase inputs as the first and second input signals of the current comparator during the auto-zero phase and to provide compare phase inputs as the first and second input signals of the current capacitor during the compare phase.

9. The DC-DC converter of claim 8, wherein the auto-zero phase inputs provided as the first and second input signals during the auto-zero phase are a same signal, and the compare phase inputs provided as the first and second input signals during the compare phase are a pair of signals being compared such that the output of the current comparator is provided based on the comparison of the pair of signals.

10. The DC-DC converter of claim 8, further comprising:

a power switch having a PMOS transistor in series with an NMOS transistor, wherein a first internal circuit node between the PMOS transistor and the NMOS transistor is configured to be coupled to an inductor.

11. The DC-DC converter of claim 10, wherein:

during the auto-zero phase, the input signal switch couples a first ground signal to each of the first and second input signals of the current comparator, and

during the compare phase, the input signal switch couples a second internal circuit node of the power switch and a second ground signal to the first and second input signals of the current comparator, wherein the output of

the current comparator indicates when a voltage on the second internal circuit node reaches a voltage of the second ground signal.

12. The DC-DC converter of claim 11, further comprising:

control circuitry, wherein when the output of the current comparator is asserted to indicate that the voltage on the second internal circuit node has reached the voltage of the second ground signal, the control circuitry disables the NMOS transistor of the power switch.

13. The DC-DC converter of claim 12, wherein:

the power switch further comprises an additional NMOS transistor coupled between the NMOS transistor of the power switch and a third voltage supply terminal configured to provide the second ground signal,

the second internal circuit node of the power switch corresponds to a node between the NMOS transistor and additional NMOS transistor of the power switch, and

the first ground signal is provided by one of the second voltage supply terminal or the third voltage supply terminal.

14. The DC-DC converter of claim 10, wherein:

during the auto-zero phase, the input signal switch couples a supply voltage to each of the first and second input signals of the current comparator, and

during the compare phase, the input signal switch couples a voltage indicative of a predetermined maximum current through the inductor and the internal circuit node of the power switch to the first and second input signals of the current comparator, wherein the output of the current comparator indicates when a voltage on the internal circuit node reaches the voltage indicative of the predetermined maximum current.

15. The DC-D converter of claim 14, further comprising:

a replica circuit configured to provide the voltage indicative of the predetermined maximum current to the input signal switch.

16. A current comparator, comprising:

a first capacitor having a first terminal coupled to a first input signal;

a second capacitor having a first terminal coupled to a second input signal;

a first transistor having a first current electrode coupled to a first voltage supply terminal via a first current source, a control electrode coupled to a second terminal of the first capacitor, and a second current electrode coupled to a first circuit node;

a second transistor having a first current electrode coupled to the first current electrode of the first transistor, a control electrode coupled to a second terminal of the second capacitor, and a second current electrode coupled to a second circuit node;

a third transistor having a first current electrode coupled to the first circuit node, a second current electrode coupled to a second voltage supply terminal, and a control electrode coupled to the first circuit node;

a fourth transistor having a first current electrode coupled to the second circuit node, a control electrode coupled to the control electrode of the third transistor, and a second current electrode coupled to the second voltage supply terminal; and

a single-ended cascode amplifier having an input coupled via a third capacitor to the second circuit node and an output,

wherein the current comparator is configured to operate in an auto-zero phase followed by a compare phase, wherein, during the auto-phase, the current comparator is configured to short the control electrode to the second current electrode of each of the first and second transistors, and during the compare phase, the current comparator is configured to not short the control electrode to the second current electrode of each of the first and second transistors.

17. The current comparator of claim **16**, further comprising:

a logic stage coupled to receive the output of the single-ended cascode amplifier, and configured to provide an output of the current comparator.

18. The current comparator of claim **16**, wherein the single-ended cascode amplifier comprises:

a fifth transistor having a control electrode configured as the input of the single-ended cascode amplifier, a first current electrode coupled to the second voltage supply terminal, and a second current electrode; and

a sixth transistor having a first current electrode coupled to the second current electrode of the fifth transistor, a control electrode, and a second current electrode coupled to the output of the single-ended cascode amplifier,

wherein the current comparator is further configured to:

during the auto-zero phase, short the control electrode of the fifth transistor to the second current electrode of the fifth transistor and short the control electrode of the sixth transistor to the second current electrode of the sixth transistor, and

during the compare phase, not short the control electrode to the second current electrode of each of the fifth and sixth transistors.

19. A DC-DC converter comprising:

the current comparator of claim **16**;

a power switch having a PMOS transistor in series with an NMOS transistor, wherein an internal circuit node between the PMOS transistor and the NMOS transistor is configured to be coupled to an inductor,

wherein:

during the auto-zero phase, a first ground signal is provided as each of the first and second input signals of the current comparator, and

during the compare phase, a second internal circuit node of the power switch and a second ground signal are provided to the first and second input signals of the current comparator, wherein the output of the current comparator indicates when a voltage on the second internal circuit node reaches a voltage of the second ground signal.

20. A DC-DC converter comprising:

the current comparator of claim **16**;

a power switch having a PMOS transistor in series with an NMOS transistor, wherein an internal circuit node between the PMOS transistor and the NMOS transistor is configured to be coupled to an inductor,

wherein:

during the auto-zero phase, a supply voltage is provided as each of the first and second input signals of the current comparator, and

during the compare phase, a voltage indicative of a predetermined maximum current through the inductor and the internal circuit node are coupled to the first and second input signals of the current comparator, wherein the output of the current comparator indicates when a voltage on the internal circuit node reaches the voltage indicative of the predetermined maximum current.

* * * * *